
EIGENVECTORS, PCA AND SVD PROBLEM SHEET

1 In this question we consider real symmetric matrices. Throughout, $\mathbf{M}$ is an $n \times n$ real symmetric matrix with eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ (repeated eigenvalues being listed according to their multiplicity).

Recall if $\mathbf{v}$ is an eigenvector then $\mathbf{M}\mathbf{v} = \lambda\mathbf{v}$ or $(\mathbf{M} - \lambda\mathbf{I})\mathbf{v} = \mathbf{0}$. As all vectors in the direction $\mathbf{v}$ are mapped to $\mathbf{0}$ the volume of space after applying $(\mathbf{M} - \lambda\mathbf{I})$ is zero. Consequently, $\det(\mathbf{M} - \lambda\mathbf{I}) = 0$. This is the characteristic equation that allows us to find λ . Using the values of λ we have found we can then solve $(\mathbf{M} - \lambda\mathbf{I})\mathbf{v} = \mathbf{0}$ for $\mathbf{v}$.

(a) Find the eigenvalues and eigenvectors of

$$\mathbf{A} = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$$

and verify that the eigenvectors are orthogonal.

The characteristic equation is $(3 - \lambda)^2 - 1 = 0$ so that $\lambda = 3 \pm 1$, giving $\lambda_1 = 4$ and $\lambda_2 = 2$. Solving $(\mathbf{A} - 4\mathbf{I})\mathbf{v} = \mathbf{0}$ gives $\mathbf{v}_1 = (1, 1)^T / \sqrt{2}$ and $(\mathbf{A} - 2\mathbf{I})\mathbf{v} = \mathbf{0}$ gives $\mathbf{v}_2 = (1, -1)^T / \sqrt{2}$. Then $\mathbf{v}_1^T \mathbf{v}_2 = (1 - 1)/2 = 0$, so they are orthogonal.

[3 marks]

(b) Show that the eigenvalues of $\mathbf{M}$ are real, and that two eigenvectors whose eigenvalues differ are necessarily orthogonal. (Hint, for the first part consider $\mathbf{v}^* \mathbf{M} \mathbf{v}$ where $\mathbf{v}^* = \bar{\mathbf{v}}^T$ is the complex-conjugate transpose.)

Let $\mathbf{M}\mathbf{v} = \lambda\mathbf{v}$ where we allow $\mathbf{v}$ and λ to be complex. Multiplying on the left by the conjugate transpose $\mathbf{v}^* = \bar{\mathbf{v}}^T$

$$\mathbf{v}^* \mathbf{M} \mathbf{v} = \lambda \mathbf{v}^* \mathbf{v} = \lambda \|\mathbf{v}\|^2.$$

Since $\mathbf{M}$ is real and symmetric the left-hand side is equal to its own complex conjugate, so $\lambda \|\mathbf{v}\|^2 = \lambda^* \|\mathbf{v}\|^2$. As $\|\mathbf{v}\|^2 > 0$ we have $\lambda = \lambda^*$, that is λ is real.

For the second part, let $\mathbf{M}\mathbf{v}_i = \lambda_i \mathbf{v}_i$ and $\mathbf{M}\mathbf{v}_j = \lambda_j \mathbf{v}_j$. Then

$$\mathbf{v}_j^T \mathbf{M} \mathbf{v}_i = \lambda_i \mathbf{v}_j^T \mathbf{v}_i, \quad \mathbf{v}_j^T \mathbf{M} \mathbf{v}_i = (\mathbf{M} \mathbf{v}_j)^T \mathbf{v}_i = \lambda_j \mathbf{v}_j^T \mathbf{v}_i$$

where we used $\mathbf{M}^T = \mathbf{M}$. Subtracting, $(\lambda_i - \lambda_j) \mathbf{v}_j^T \mathbf{v}_i = 0$, so if $\lambda_i \neq \lambda_j$ then $\mathbf{v}_j^T \mathbf{v}_i = 0$.

[4 marks]

- (c) Let $\mathbf{V}$ be the matrix whose columns are the normalised eigenvectors of $\mathbf{M}$ and let $\mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_n)$. Explain why $\mathbf{V}$ is orthogonal, show that $\mathbf{M} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^T$, and deduce an expression for $\mathbf{M}^k$.

By the previous part the eigenvectors are mutually orthogonal and we have normalised them, so $\mathbf{V}^T\mathbf{V} = \mathbf{I}$, that is $\mathbf{V}$ is orthogonal and $\mathbf{V}^{-1} = \mathbf{V}^T$. (For a repeated eigenvalue we may always choose an orthonormal basis of the corresponding eigenspace.) The n eigenvector equations written together are $\mathbf{M}\mathbf{V} = \mathbf{V}\mathbf{\Lambda}$, so right-multiplying by $\mathbf{V}^{-1} = \mathbf{V}^T$

$$\mathbf{M} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^T.$$

Then $\mathbf{M}^2 = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^T\mathbf{V}\mathbf{\Lambda}\mathbf{V}^T = \mathbf{V}\mathbf{\Lambda}^2\mathbf{V}^T$ since $\mathbf{V}^T\mathbf{V} = \mathbf{I}$, and by induction $\mathbf{M}^k = \mathbf{V}\mathbf{\Lambda}^k\mathbf{V}^T$.

[3 marks]

- (d) Using $\det(\mathbf{AB}) = \det(\mathbf{A})\det(\mathbf{B})$, show that

$$\det(\mathbf{M}) = \prod_{i=1}^n \lambda_i.$$

From the previous part

$$\det(\mathbf{M}) = \det(\mathbf{V}\mathbf{\Lambda}\mathbf{V}^T) = \det(\mathbf{V})\det(\mathbf{\Lambda})\det(\mathbf{V}^T).$$

But $\det(\mathbf{V})\det(\mathbf{V}^T) = \det(\mathbf{V}\mathbf{V}^T) = \det(\mathbf{I}) = 1$, so $\det(\mathbf{M}) = \det(\mathbf{\Lambda})$. The determinant of a diagonal matrix is the product of its diagonal elements, hence $\det(\mathbf{M}) = \prod_i \lambda_i$.

[5 marks]

- (e) Using the cyclic property of the trace, $\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$, show that

$$\text{tr} \mathbf{M} = \sum_{i=1}^n \lambda_i.$$

Writing $\mathbf{M} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^T$ and treating $\mathbf{A} = \mathbf{V}$ and $\mathbf{B} = \mathbf{\Lambda}\mathbf{V}^T$,

$$\text{tr} \mathbf{M} = \text{tr}(\mathbf{V}\mathbf{\Lambda}\mathbf{V}^T) = \text{tr}(\mathbf{\Lambda}\mathbf{V}^T\mathbf{V}) = \text{tr} \mathbf{\Lambda} = \sum_{i=1}^n \lambda_i$$

where we used $\mathbf{V}^T\mathbf{V} = \mathbf{I}$.

[5 marks]

End of question 1

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2 Let $\mathbf{X}$ be a $d \times m$ matrix whose m columns are (mean-subtracted) data vectors of dimension d , scaled so that the covariance matrix is $\mathbf{C} = \mathbf{X}\mathbf{X}^\top$. We also define the dual matrix $\mathbf{D} = \mathbf{X}^\top\mathbf{X}$.

- (a) Show that $\mathbf{C}$ is symmetric and positive semi-definite. Under what condition is it positive definite?

$\mathbf{C}^\top = (\mathbf{X}\mathbf{X}^\top)^\top = \mathbf{X}\mathbf{X}^\top = \mathbf{C}$, so it is symmetric. For any $\mathbf{w}$

$$\mathbf{w}^\top \mathbf{C} \mathbf{w} = \mathbf{w}^\top \mathbf{X} \mathbf{X}^\top \mathbf{w} = (\mathbf{X}^\top \mathbf{w})^\top (\mathbf{X}^\top \mathbf{w}) = \|\mathbf{X}^\top \mathbf{w}\|^2 \geq 0$$

so $\mathbf{C}$ is positive semi-definite. It is positive definite if $\mathbf{X}^\top \mathbf{w} \neq \mathbf{0}$ for every $\mathbf{w} \neq \mathbf{0}$, that is when the rows of $\mathbf{X}$ are linearly independent, so $\mathbf{X}$ has rank d . This requires at least d data points, $m \geq d$.

[3 marks]

- (b) The first principal component is the unit vector $\mathbf{w}$ maximising the variance $\mathbf{w}^\top \mathbf{C} \mathbf{w}$ of the projected data. Using a Lagrange multiplier to impose $\|\mathbf{w}\| = 1$, show that $\mathbf{w}$ must be an eigenvector of $\mathbf{C}$, and identify the value of the Lagrange multiplier.

The Lagrangian is

$$L(\mathbf{w}, \lambda) = \mathbf{w}^\top \mathbf{C} \mathbf{w} - \lambda(\mathbf{w}^\top \mathbf{w} - 1).$$

Since $\mathbf{C}$ is symmetric, $\nabla_{\mathbf{w}} L = 2\mathbf{C}\mathbf{w} - 2\lambda\mathbf{w}$, and setting this to zero gives

$$\mathbf{C}\mathbf{w} = \lambda\mathbf{w},$$

so $\mathbf{w}$ is an eigenvector of $\mathbf{C}$. Multiplying on the left by $\mathbf{w}^\top$ and using $\mathbf{w}^\top \mathbf{w} = 1$,

$$\mathbf{w}^\top \mathbf{C} \mathbf{w} = \lambda.$$

That is, the Lagrange multiplier is the variance along $\mathbf{w}$, so to maximise the variance we choose the eigenvector with the largest eigenvalue.

[5 marks]

- (c) Let $\mathbf{v}_1, \dots, \mathbf{v}_d$ be the normalised eigenvectors of $\mathbf{C}$ ordered by decreasing eigenvalue and let $\mathbf{P} = \sum_{i=1}^k \mathbf{v}_i \mathbf{v}_i^\top$. Show that $\mathbf{P}^2 = \mathbf{P}$ and that the expected squared reconstruction error $\mathbb{E}[\|\mathbf{P}\mathbf{x} - \mathbf{x}\|^2]$ is $\sum_{i>k} \lambda_i$.

Using $\mathbf{v}_i^\top \mathbf{v}_j = \delta_{ij}$,

$$\mathbf{P}^2 = \sum_{i=1}^k \sum_{j=1}^k \mathbf{v}_i (\mathbf{v}_i^\top \mathbf{v}_j) \mathbf{v}_j^\top = \sum_{i=1}^k \mathbf{v}_i \mathbf{v}_i^\top = \mathbf{P}.$$

Expanding $\mathbf{x}$ in the complete eigenbasis, $\mathbf{x} = \sum_{i=1}^d (\mathbf{v}_i^\top \mathbf{x}) \mathbf{v}_i$, so $\mathbf{P}\mathbf{x} - \mathbf{x} = -\sum_{i>k} (\mathbf{v}_i^\top \mathbf{x}) \mathbf{v}_i$ and, by orthonormality,

$$\mathbb{E}[\|\mathbf{P}\mathbf{x} - \mathbf{x}\|^2] = \sum_{i>k} \mathbb{E}[(\mathbf{v}_i^\top \mathbf{x})^2] = \sum_{i>k} \mathbf{v}_i^\top \mathbf{C} \mathbf{v}_i = \sum_{i>k} \lambda_i.$$

[3 marks]

- (d) Show that if v is an eigenvector of $\mathbf{D}$ with non-zero eigenvalue λ then $\mathbf{X}v$ is an eigenvector of $\mathbf{C}$ with the same eigenvalue. Why must we exclude $\lambda = 0$? Why is this result useful when there are many more features than examples?

Acting with $\mathbf{C}$ on $\mathbf{X}v$,

$$\mathbf{C}(\mathbf{X}v) = \mathbf{X}\mathbf{X}^T\mathbf{X}v = \mathbf{X}(\mathbf{D}v) = \lambda\mathbf{X}v$$

so $\mathbf{X}v$ is an eigenvector of $\mathbf{C}$ with eigenvalue λ , provided $\mathbf{X}v \neq \mathbf{0}$. Now

$$\|\mathbf{X}v\|^2 = v^T\mathbf{X}^T\mathbf{X}v = v^T\mathbf{D}v = \lambda\|v\|^2$$

so $\mathbf{X}v \neq \mathbf{0}$ exactly when $\lambda \neq 0$; for $\lambda = 0$ the construction gives the zero vector, which is not an eigenvector.

$\mathbf{C}$ is $d \times d$ and $\mathbf{D}$ is $m \times m$. When $d \gg m$ it is far cheaper to find the eigensystem of $\mathbf{D}$ (cost $O(m^3)$) and map the eigenvectors across than to work with $\mathbf{C}$ directly (cost $O(d^3)$). Note that $\mathbf{C}$ then has at most m non-zero eigenvalues: the data cannot span the feature space.

[5 marks]

- (e) The singular value decomposition writes $\mathbf{X} = \mathbf{U}\mathbf{S}\mathbf{V}^T$, where $\mathbf{U}$ and $\mathbf{V}$ are orthogonal and $\mathbf{S}$ is diagonal with non-negative entries s_i . Show that

$$\mathbf{X}\mathbf{X}^T = \mathbf{U}\mathbf{S}\mathbf{S}^T\mathbf{U}^T, \quad \mathbf{X}^T\mathbf{X} = \mathbf{V}\mathbf{S}^T\mathbf{S}\mathbf{V}^T$$

and hence say what the columns of $\mathbf{U}$ and $\mathbf{V}$ are, and how the singular values relate to the eigenvalues.

Substituting the decomposition and using $\mathbf{V}^T\mathbf{V} = \mathbf{I}$,

$$\mathbf{X}\mathbf{X}^T = \mathbf{U}\mathbf{S}\mathbf{V}^T\mathbf{V}\mathbf{S}^T\mathbf{U}^T = \mathbf{U}\mathbf{S}\mathbf{S}^T\mathbf{U}^T$$

and similarly, using $\mathbf{U}^T\mathbf{U} = \mathbf{I}$,

$$\mathbf{X}^T\mathbf{X} = \mathbf{V}\mathbf{S}^T\mathbf{U}^T\mathbf{U}\mathbf{S}\mathbf{V}^T = \mathbf{V}\mathbf{S}^T\mathbf{S}\mathbf{V}^T.$$

Both right-hand sides have the form $\mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^T$ with $\mathbf{Q}$ orthogonal and $\mathbf{\Lambda}$ diagonal, which by question 1(c) is the eigendecomposition. So the columns of $\mathbf{U}$ are the eigenvectors of $\mathbf{X}\mathbf{X}^T$, the columns of $\mathbf{V}$ are the eigenvectors of $\mathbf{X}^T\mathbf{X}$, and the non-zero eigenvalues of both are s_i^2 . Equivalently the singular values are the square roots of the non-zero eigenvalues, which is the duality of part (d) again.

[4 marks]

End of question 2

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3 In this question I want you to use a numerical package to check the results of the previous two questions. You can use MATLAB, OCTAVE, numpy in Python or any other language of your choosing. The answers are only going to be accurate up to the numerical precision of your machine (typically around 10^{-16}), so two quantities should be treated as equal when they agree to around ten decimal places. For python the following may be useful

<code>import numpy as np</code>	import the numpy module
<code>A = np.random.randn(n,m)</code>	generate a matrix of normally distributed random deviates with n rows and m columns
<code>A.transpose()</code>	return the transpose of A (does not change A)
<code>A@B</code>	matrix multiplication AB . Beware that <code>A*B</code> performs elementwise multiplication which is different to normal matrix multiplication
<code>A.trace()</code>	returns the trace of a matrix
<code>np.linalg.det(A)</code>	return the determinant of a matrix
<code>np.linalg.eigh(A)</code>	eigenvalues and eigenvectors of a <i>symmetric</i> matrix A , returned as a pair (<code>lam</code> , <code>V</code>) where <code>lam</code> is a one-dimensional array of eigenvalues in increasing order and the columns of <code>V</code> are the corresponding normalised eigenvectors
<code>np.linalg.eig(A)</code>	the same for a general square matrix. The eigenvalues come in no particular order and may be complex
<code>np.linalg.svd(X)</code>	singular value decomposition, returned as a triple (<code>U</code> , <code>s</code> , <code>Vt</code>). Note that <code>s</code> is a one-dimensional array of singular values in decreasing order, not a matrix, and that the third element returned is $\mathbf{V}^T$ and not V
<code>np.diag(s)</code>	turn a one-dimensional array into a diagonal matrix (and, given a matrix, extract its diagonal)
<code>np.sort(x)</code>	sort an array into increasing order
<code>np.allclose(a, b)</code>	test whether two arrays agree to within numerical precision. Returns a single true or false value
<code>np.round(A, 10)</code>	rounds the elements in a matrix A to 10 decimal places. Makes expressions like <code>(A@B)@C - A@(B@C)</code> easier to read

Generate a random 5×5 matrix **M** and form the symmetric matrix $\mathbf{A} = \mathbf{M} + \mathbf{M}^T$. Separately generate a random 4×6 matrix **X**.

- (a) Compute the eigenvalues of **A**. Compare the product of the eigenvalues with `np.linalg.det(A)`, and their sum with `A.trace()`. Which of the results of question 1 does this check?

This checks $\det(\mathbf{M}) = \prod_i \lambda_i$ and $\text{tr } \mathbf{M} = \sum_i \lambda_i$ from question 1(d) and 1(e).

```
M = np.random.randn(5,5)
A = M + M.transpose()
lam, V = np.linalg.eigh(A)
print(np.prod(lam), np.linalg.det(A))
print(np.sum(lam), A.trace())
```

The determinant is the more delicate comparison: it is a product of five numbers so the relative errors accumulate.

[3 marks]

- (b) Verify numerically that $\mathbf{V}^T \mathbf{V} = \mathbf{I}$ and that $\mathbf{A} = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^T$, where $\mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_5)$. What happens if you replace $\mathbf{A}$ by the non-symmetric $\mathbf{M}$?

```
print(np.round(V.transpose()@V, 10))
L = np.diag(lam)
print(np.round(V@L@V.transpose() - A, 10))
```

Both should print matrices of zeros (the first after subtracting the identity). For $\mathbf{M}$ the eigenvectors are not orthogonal, so $\mathbf{V}^T \mathbf{V} \neq \mathbf{I}$ and the reconstruction fails; one must use $\mathbf{V}^{-1}$ instead of $\mathbf{V}^T$. This is explored in question 4.

[3 marks]

- (c) Compute the singular value decomposition of $\mathbf{X}$. Check that the singular values are the square roots of the non-zero eigenvalues of both $\mathbf{X} \mathbf{X}^T$ and $\mathbf{X}^T \mathbf{X}$.

```
X = np.random.randn(4, 6)
U, s, Vt = np.linalg.svd(X)
lc, Uc = np.linalg.eigh(X@X.transpose())
ld, Vd = np.linalg.eigh(X.transpose()@X)
print(np.sort(s**2))
print(np.round(lc, 10))
print(np.round(ld, 10))
```

The four values of s_i^2 match the four eigenvalues of $\mathbf{X} \mathbf{X}^T$ and the four largest of the six eigenvalues of $\mathbf{X}^T \mathbf{X}$; the remaining two are zero. Note that `eigh` returns increasing eigenvalues while `svd` returns decreasing singular values, so one list must be reversed before comparing.

[4 marks]

- (d) Check that the columns of $\mathbf{U}$ are eigenvectors of $\mathbf{X} \mathbf{X}^T$ and the columns of $\mathbf{V}$ are eigenvectors of $\mathbf{X}^T \mathbf{X}$. You will probably find that some columns come back with the opposite sign to the ones you computed in the previous part. Why is this not an error?

If $\mathbf{A} \mathbf{v} = \lambda \mathbf{v}$ then $\mathbf{A}(-\mathbf{v}) = \lambda(-\mathbf{v})$, so $-\mathbf{v}$ is just as good an eigenvector and normalising to unit length does not resolve the choice. Which one a routine returns depends on the algorithm, so the sign carries no meaning. (If an eigenvalue is repeated, the ambiguity is larger still: any orthonormal basis of that eigenspace will do.) A safe check compares directions rather than vectors,

```
for i in range(4):
    c = U[:, i]@Uc[:, 3-i]
    print(np.round(abs(c), 10))
```

each of which should print 1. Equivalently, compare $\mathbf{u} \mathbf{u}^T$, which is insensitive to the sign.

[4 marks]

- (e) $\mathbf{X}$ is 4×6 , so one of $\mathbf{X} \mathbf{X}^T$ and $\mathbf{X}^T \mathbf{X}$ is necessarily singular. Which, and why? Confirm this from the eigenvalues and from the determinant.

$\mathbf{X}$ has rank at most $\min(4, 6) = 4$, and both products have the same rank as $\mathbf{X}$. $\mathbf{X} \mathbf{X}^T$ is 4×4 and generically has full rank, so it is invertible. $\mathbf{X}^T \mathbf{X}$ is 6×6 with rank at most 4, so at least two of its eigenvalues are zero and its determinant vanishes. This is the statement of question 2(d) seen from the other side: the two matrices share their non-zero eigenvalues, and the larger one makes up the difference with zeros.

[3 marks]

- (f) Form the rank- k approximation $\mathbf{X}_k = \sum_{i=1}^k s_i \mathbf{u}_i \mathbf{v}_i^T$ for $k = 1, 2, 3, 4$ and compare the squared error $\sum_{jl} (X_{jl} - (X_k)_{jl})^2$ with $\sum_{i>k} s_i^2$.

```
for k in range(1,5):
    Xk = U[:, :k]@np.diag(s[:k])@Vt[:, :]
    err = ((X-Xk)**2).sum()
    print(err, (s[k:]**2).sum())
```

The two agree for every k , and the error falls to zero at $k = 4$ because $\mathbf{X}$ has rank 4. This is the matrix version of the PCA reconstruction result of question 2(c): the error you incur is the sum of the eigenvalues you throw away.

[3 marks]

End of question 3

- Do not write in this space •

4 Almost everything we have done so far assumed the matrix was real and symmetric, which guaranteed real eigenvalues and a complete set of orthogonal eigenvectors. This question is about what happens when that assumption is dropped. The point is not that the calculations are harder, but that the conclusions genuinely change. You may use python throughout.

(a) By hand, find the eigenvalues and eigenvectors of

$$\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Show that $\mathbf{A}$ has a repeated eigenvalue but only one eigenvector direction, and explain why this means $\mathbf{A}$ cannot be written as $\mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1}$.

The matrix is upper triangular so the characteristic equation is $(1 - \lambda)^2 = 0$, giving $\lambda = 1$ twice. Solving $(\mathbf{A} - \mathbf{I})\mathbf{v} = \mathbf{0}$,

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow v_2 = 0$$

so every eigenvector is a multiple of $(1, 0)^T$: there is only one eigenvector direction, not two. To write $\mathbf{A} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1}$ the columns of $\mathbf{V}$ must be n linearly independent eigenvectors, so that $\mathbf{V}^{-1}$ exists. Here there are not enough, so no such $\mathbf{V}$ exists. A matrix like this is called *defective*.

[4 marks]

(b) Confirm this in python using `np.linalg.eig`. You will get back a matrix with two columns: look at them and explain what has happened. Compute the determinant of that matrix and say why $\mathbf{V}^{-1}$ cannot be formed.

```
A = np.array([[1.0, 1.0], [0.0, 1.0]])
lam, V = np.linalg.eig(A)
print(lam)
print(V)
print(np.linalg.det(V))
```

The routine always returns an $n \times n$ matrix, but here both columns are the same vector $(1, 0)^T$ up to sign and rounding, because that is the only eigenvector there is. The columns are therefore linearly dependent, $\det(\mathbf{V}) \approx 0$, and $\mathbf{V}$ is singular so $\mathbf{V}^{-1}$ does not exist. `np.linalg.eig` gives no warning: it is the user's job to notice.

[3 marks]

(c) Now take

$$\mathbf{B} = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}.$$

Find its eigenvectors and show that, although they are not orthogonal, $\mathbf{B} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1}$ still holds. What does this tell you about the difference between *non-symmetric* and *defective*?

The eigenvalues are $\lambda_1 = 1$ and $\lambda_2 = 2$. For $\lambda_1 = 1$, $(\mathbf{B} - \mathbf{I})\mathbf{v} = \mathbf{0}$ gives $v_2 = 0$ so $\mathbf{v}_1 = (1, 0)^T$. For $\lambda_2 = 2$, $(\mathbf{B} - 2\mathbf{I})\mathbf{v} = \mathbf{0}$ gives $-v_1 + v_2 = 0$ so $\mathbf{v}_2 = (1, 1)^T/\sqrt{2}$. These are independent, so $\mathbf{V}$ is invertible and the decomposition holds, but $\mathbf{v}_1^T \mathbf{v}_2 = 1/\sqrt{2} \neq 0$, so they are not orthogonal and $\mathbf{V}^T \mathbf{V} \neq \mathbf{I}$. In python `np.linalg.inv(V) @ B @ V` recovers $\mathbf{\Lambda}$ while `V.transpose() @ B @ V` does not. So the two properties are separate. Symmetry buys *orthogonal* eigenvectors; what diagonalisation actually needs is only that there are n independent ones. Most non-symmetric matrices are diagonalisable; defective ones like $\mathbf{A}$ are the exception.

[3 marks]

(d) Consider the rotation matrix

$$\mathbf{R} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}.$$

Compute its eigensystem numerically for $\theta = \pi/3$. Verify that the eigenvalues are $e^{\pm i\theta}$ and that the two eigenvectors are complex conjugates of each other. Explain why complex eigenvalues of a *real* matrix must come in conjugate pairs. What replaces the sign ambiguity of question 3(d) in the complex case?

The characteristic equation is $\lambda^2 - 2\cos\theta\lambda + 1 = 0$, with roots $\lambda = \cos\theta \pm i\sin\theta = e^{\pm i\theta}$, and the eigenvectors are $(1, \mp i)^T/\sqrt{2}$. Geometrically this is what we should expect: a rotation leaves no real direction unchanged (for θ not a multiple of π), so there is no real eigenvector.

The characteristic polynomial of a real matrix has real coefficients, so if $p(\lambda) = 0$ then taking the complex conjugate of the whole equation gives $p(\lambda^*) = 0$ also. Likewise conjugating $\mathbf{R}\mathbf{v} = \lambda\mathbf{v}$ gives $\mathbf{R}\mathbf{v}^* = \lambda^*\mathbf{v}^*$, so the eigenvectors pair up in the same way. In question 3(d) an eigenvector was fixed only up to a factor of ± 1 . For a complex eigenvector it is fixed only up to an arbitrary phase $e^{i\phi}$, since $\|e^{i\phi}\mathbf{v}\| = \|\mathbf{v}\|$. The real case is just $\phi \in \{0, \pi\}$.

[4 marks]

(e) Check that $\det(\mathbf{R}) = \prod_i \lambda_i$ and $\text{tr } \mathbf{R} = \sum_i \lambda_i$ still hold, even though the eigenvalues are complex. Why do the answers come out real?

$\det(\mathbf{R}) = \cos^2\theta + \sin^2\theta = 1$, and $\prod_i \lambda_i = e^{i\theta}e^{-i\theta} = 1$. $\text{tr } \mathbf{R} = 2\cos\theta$, and $\sum_i \lambda_i = e^{i\theta} + e^{-i\theta} = 2\cos\theta$. Both results hold for any square matrix, not just symmetric ones, because they follow from the characteristic polynomial rather than from any orthogonality. The answers are real because the complex eigenvalues occur in conjugate pairs, and both the sum and the product of a conjugate pair are real.

[3 marks]

- (f) Finally, compute the singular value decomposition of the defective matrix $\mathbf{A}$ of part (a). It exists, and its singular values are real and positive, even though $\mathbf{A}$ has no eigendecomposition. Explain why, and confirm that $s_1 s_2 = |\det(\mathbf{A})|$.

The SVD of $\mathbf{A}$ is built from the eigendecompositions of $\mathbf{A}^T \mathbf{A}$ and $\mathbf{A} \mathbf{A}^T$, which are symmetric and positive semi-definite whatever $\mathbf{A}$ is. So an SVD exists for *every* matrix, square or not, defective or not. Here $\mathbf{A}^T \mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$, with eigenvalues $(3 \pm \sqrt{5})/2$ and hence $s_1 \approx 1.618$, $s_2 \approx 0.618$. Then $s_1 s_2 = 1 = |\det(\mathbf{A})|$, as it must be since $|\det(\mathbf{A})| = |\det(\mathbf{U})| |\det(\mathbf{S})| |\det(\mathbf{V}^T)| = \prod_i s_i$. The matrix is invertible: its determinant is non-zero and no singular value is zero. Being defective is a statement about eigenvectors, not about invertibility.

[3 marks]

End of question 4